Research Statement

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My research to date spans three main areas: Bayesian inference for spatial statistics, surrogate modeling for computer experiments, and inferential models for decision making theory. My work in the first two areas was motivated by real-world applications and contributed to the fields by extending methods to multivariate settings, be it multiple outcomes or multiple computer experiments. My current project in the third area develops decision-making theory using possibilistic inferential models (IM), a framework outside the likelihood-based methods and Bayesian framework, to provide theoretical guarantees. An overarching link of the three projects is the consideration, utility, and calibration of uncertainty quantification (UQ). After graduation, I plan to continue examining UQ and balancing applications and theoretical development.

1 Bayesian inference for spatial statistics

Pollutants, such as Per- and Polyfluoroalkyl Substances (PFAS), are often co-existing, correlated, and causing issues on multiple conditions. Furthermore, it is often difficult to know if all confounders have been included in the model. Using causal inference in the context of spatial data has received increasing attention in recent years (Reich et al., 2021). These reasons motivated my advisor, Dr. Brian Reich, his former postdoctoral fellow, Dr. Yawen Guan, and me to develop a multivariate spectral model for multiple exposures and multiple outcomes. Originated from Guan et al. (2023), the method first projects the spatial information into the spectral domain. The immediate benefit of the spectral projection is that, since the information is now independent across frequencies, computation is fast.

Before a regression model can be constructed for the spectral information, two challenges emerge: 1) unmeasured spatial confounding is likely in environmental applications. Thus, we relax the no-unmeasured-confounder assumption in causal inference and allow for confounding in global scales (i.e. frequencies corresponding to small eigenvalues). Each spatial scale thus receives its own coefficient estimate. In other words, the model provides multiscale coefficient estimates, and the estimates from spatial scales deemed unconfounded will be unbiased. This, however, leads to the second challenge: 2) the multiscale coefficient estimates for multiple exposures and multiple outcomes are in a three-way tensor and high-dimensional. To reduce dimension, we utilize canonical polyadic (CP) tensor decomposition, ensuring efficient computation. I constructed the fully Bayesian model with several MCMC samplers in R, and the manuscript was submitted in June 2025 and is currently under review (Prim et al., 2025).

While the multiscale estimates from our model are found to have low bias, an added benefit is that the credible intervals from our proposed method are more conservative, since the model essentially removes the information that comes from confounded spatial scales. The wider credible intervals have nominal or higher coverage rates, while spatial (i.e. single-scale) or non-spatial models can have coverage rates much lower than the nominal level. This highlights the importance of considering both point estimation and uncertainty quantification (UQ) capabilities of a model. Even though all fully Bayesian models provide posterior distributions and convenient UQ, not all provide correctly calibrated UQ and it is up to the researchers to carefully consider issues around UQ.

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1.1 Future directions

I plan to continue the work in spatial statistics and study the topic of unmeasured spatial confounding. Methods that account for unmeasured spatial confounding sometimes lead to unintuitive or contradictory results (Dupont et al., 2024). The role played by spatial scales is particular central; confounding can vary at different scales and paying close attention to the scales can help disentangle different degrees of confounding (Paciorek, 2010).

Further, my experience in spatial analysis can potentially transfer to image analysis, e.g. machine learning for MRI imaging analysis. I will seek collaboration opportunities with medical researchers, such as Mike Chan, Professor of Radiation Oncology at Wake Forest University School of Medicine, with whom I have previously collaborated.

2 Surrogate modeling for computer experiment

My second research project started in my 2024 summer internship at the Lawrence Livermore National Laboratory (LLNL), which was centered around flight dynamics. Simply put, aerodynamics throughout a trajectory depends on many design variables, e.g. speed, altitude, etc. High fidelity computational fluid dynamics is often infeasible, and thus surrogate modeling has been used to rapidly generate predictions to construct aerodynamic databases instead of relying on computer experiments (Gramacy, 2020; Sauer, 2023). Further, if only certain regions of the surface (failure or success region), more efficient ways to choose the design points can be used. To meet this goal, my LLNL mentor (Dr. Kevin Quinlan), Dr. Annie Booth (a NCSU faculty at that time but currently at Virginia Tech who has worked extensively in this particular area), and I developed a method utilizing active learning and contour location to work simultaneously for multiple computer experiments (Quinlan et al., 2024).

Specifically, an algorithm is needed to efficiently find an optimal design point that leads to multiple responses that meet a certain criterion. Specifically, let $f^{(r)}: \mathcal{X} \to \mathbb{R}$ denote the r^{th} computer experiment for $r = 1, \ldots, R$. Each $f^{(r)}$ is independent but acts on the same domain $\mathcal{X} \subset \mathbb{R}^d$. Our goal is to identify the "optimal design point":

$$\left\{ \tilde{\mathbf{x}} \in \mathcal{X} \;\middle|\; f^{(r)}(\tilde{\mathbf{x}}) = \tau_r \;\; \forall \;\; r = 1, \cdots, R \right\}, \tag{1}$$

where τ_r are pre-specified target response values. One major challenge is that current contour location methods are for one single computer experiment (Booth et al., 2025; Cole et al., 2023). Since then, we developed a method that maximizes the predictive probability of being close enough to all contours to target the intersection of multiple contours. The use of predictive probabilities highlights the utility of UQ; the effectiveness of the method hinges on well-calibrated UQ. To avoid wasting the sampling budget when UQ is not yet correct—which happens where not enough design points have been acquired—we sample points from the Pareto front to explore areas with high uncertainty. Our method—joint Contour Location (jCL)—balances exploitation and exploration and outperforms competitors in both simulations and a real-world application. We are currently completing the manuscript (Prim et al., 2025+) and planning to submit the paper and upload a preprint this November.

2.1 Future directions

While the paper uses a real-world application, it is a simplified example to prove the effectiveness of jCL, in which we assume one unique optimal design point exists. For the next stage, we hope to explore more difficult scenarios, in which no or multiple optimal design points exist. Further, I have always been interested in software development. After the paper is submitted, I plan to package the Python code into a Python package, allowing for easy installation and use.

3 Decision making using possibilistic inferential models (IM)

While developing methods for tangible applications is exciting, I am also deeply interested in fundamentals and theoretical guarantees of valid uncertainty quantification. After two projects that consider and utilize UQ capabilities, I turned to possibilistic inferential models (IM), a framework that provides UQ and guarantees error rates to be below a prespecified level (Martin, 2025c). Based on possibilistic measures, IM incorporates imprecise probability and considers worst case scenarios. Optimization—rather than integration—is a central tool of IM and requires expensive computation. Some new strategies based on Monte Carlo sampling have been proposed and opened up exciting opportunities (Cella and Martin, 2025; Martin, 2025a).

With my two other co-advisors, Dr. Ryan Martin and Dr. Jonathan Williams, I am currently exploring how to apply IM on decision making theory, extending the foundation laid in Martin (2025b). We are examining different examples to demonstrate the strengths of using IM in decision theory and exploring efficient computation strategies, striving to better explain how IM can offer a valid and attainable option. Our goal is to complete an invited paper for the International Journal of Approximate Reasoning (IJAR) in December 2025 and continue exploring the utility of IM on decision-making beyond this year.

3.1 Future directions

After the completion of the current decision-making project, I hope to continue working with Ryan Martin and Jonathan Williams and potentially Leo Cella at Wake Forest University to dive deeper into the UQ capabilities and validity offered by possibilistic inferential models. With faster computational strategies, it is plausible that IM can be used on more real-world application, providing well-calibrated UQ for applications that can incur huge loss from small UQ errors, such as satellite conjunction analysis (Balch et al., 2018) and air traffic management.

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